

3. HTML 5 and Cascading Style Sheets, Types of CSS

a. Write a HTML program, that makes use of <article>, <aside>, <figure>, <figcaption>, <footer>, <header>, <main>, <nav>, <section>, <div>, tags.

DESCRIPTION:

- HTML5 introduces semantic elements to clearly describe the structure and meaning of a web page. In this program, different HTML5 semantic tags are used to organize the webpage in a meaningful way.
- The <header> tag is used to display the title and introductory information of the webpage. The <nav> tag contains navigation links that help users move between different sections of the website. The <main> tag holds the main content of the webpage.
- The <section> tag is used to group related content, while the <article> tag represents independent content such as an article or blog post. The <aside> tag is used to display related information that is placed beside the main content.
- The <figure> tag is used to display an image, and the <figcaption> tag provides a caption for the image. The <div> tag is used to group multiple elements together for layout purposes. The tag is used for inline text styling within a paragraph.
- Finally, the <footer> tag is used to display footer information such as copyright details.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML5 Tag Demonstration</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 0;
    }
    header {
      background-color: #007BFF;
      color: white;
      padding: 1rem;
      text-align: center;
    }
    nav {
      background-color: #f4f4f4;
      padding: 1rem;
      text-align: center;
    }
    nav a {
      margin: 0 10px;
      text-decoration: none;
      color: #333;
    }
    main {
      margin: 1rem;
    }
    div.row {
      display: flex;
    }
    article, aside {
      border: 1px solid #ddd;
      padding: 1rem;
      margin: 1rem;
```

```

        flex: 2;
        min-width: 300px;
    }
    aside {
        flex: 1;
    }
    figure {
        margin: 0;
        text-align: center;
        width: -webkit-fill-available;
    }
    figcaption {
        font-size: 0.9rem;
        color: #555;
    }
    img {
        width: -webkit-fill-available;
    }
    footer {
        background-color: #333;
        color: white;
        text-align: center;
        padding: 1rem;
        margin-top: 1rem;
    }
</style>
</head>
<body>
    <header>
        <h1>Welcome to My HTML5 Demo Page</h1>
    </header>

    <nav>
        <a href="#home">Home</a>
        <a href="#about">About</a>
        <a href="#contact">Contact</a>
    </nav>

    <main>
        <section>
            <h2>About This Page</h2>
            <p>This page demonstrates the use of various HTML5 tags such as <span
style="color:blue;">header</span>, <span style="color:blue;">article</span>, <span
style="color:blue;">figure</span>, and others.</p>
        </section>

        <div class="row">
            <article>
                <h2>Article Section</h2>
                <p>An article typically contains standalone content like a blog post or a news article.</p>
                <figure>
                    
                    <figcaption>Figure: A humming bird</figcaption>
                </figure>
            </article>

```

```

<aside>
  <h2>Related Links</h2>
  <ul>
    <li><a href="https://www.tutorialkart.com/html/">HTML Tutorials</a></li>
    <li><a href="https://www.tutorialkart.com/css/">CSS Tutorials</a></li>
    <li><a href="https://www.tutorialkart.com/javascript/">JavaScript Basics</a></li>
  </ul>
</aside>
</div>
</main>

<footer>
  <p>&copy; 2024 HTML5 Demo Page. All rights reserved.</p>
</footer>
</body>
</html>

```

b. Write a HTML program, to embed audio and video into HTML web page.

DESCRIPTION:

- HTML5 provides multimedia elements to embed audio and video directly into a web page without using external plugins. In this program, the <audio> tag is used to embed an audio file and the <video> tag is used to embed a video file.
- The controls attribute is used to display play, pause, and volume controls to the user. The <source> tag specifies the media file and its format. If the browser does not support the audio or video format, a fallback message is displayed.

CODE:

```

<!DOCTYPE html>
<html>
<head>
  <title>Embed Audio and Video</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 20px;
      line-height: 1.6;
    }
    h1, h2 {
      color: #333;
    }
    .container {
      max-width: 800px;
      margin: 0 auto;
      text-align: center;
    }
    .video-container {
      margin: 20px 0;
    }
    .audio-container {
      margin: 20px 0;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>Embed Audio and Video in HTML</h1>

```

```

<!-- Embedding Audio -->
<div class="audio-container">
  <h2>Audio Example</h2>
  <p>Listen to this sample audio:</p>
  <audio controls>
    <source src="https://www.tutorialkart.com/wp-content/uploads/2024/12/sample-audio.mp3"
type="audio/mpeg">
    Your browser does not support the audio element.
  </audio>
<!-- Embedding Video -->
</div>
  <h2>Video Example</h2>
  <video width="400" height="250" controls>
    <source src="D:\ACET\HTML\images\video.mp4" type="video/mp4">
    Your browser does not support the video element.
  </video>

<!-- Embedding YouTube Video -->
<div class="video-container">
  <h2>YouTube Video Example</h2>
  <p>Watch this embedded YouTube video:</p>
  <iframe width="560" height="315"
src="https://www.youtube.com/embed/it1rTvBcfRg?si=lgWAdiI8cmpxr6AH&controls=0"
title="YouTube video player" frameborder="0" allow="accelerometer; autoplay; clipboard-write;
encrypted-media; gyroscope; picture-in-picture; web-share" allowfullscreen></iframe>
  </div>
</div>
</body>
</html>

```

c. Write a program to apply different types (or levels of styles or style specification formats)- inline, internal, external styles to HTML elements. (identify selector, property and value).

DESCRIPTION:

- CSS (Cascading Style Sheets) is used to control the presentation and appearance of HTML elements. In this program, three different types of style specification formats are applied to HTML elements: **inline**, **internal**, and **external** styles.
- **Inline styles** are applied directly within an HTML element using the style attribute. This style has the highest priority and affects only the specific element to which it is applied.
- **Internal styles** are defined inside the <style> tag within the <head> section of the HTML document. These styles are applied to the entire page and use **selectors** to select HTML elements.
- **External styles** are written in a separate CSS file and linked to the HTML document using the <link> tag. This method allows the same styles to be reused across multiple web pages.
- In CSS, a **selector** identifies the HTML element to be styled, a **property** defines what aspect of the element is being styled (such as color, font-size, or margin), and a **value** specifies the setting for that property.

CODE:

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CSS Styling Demonstration</title>

  <!-- Linking External CSS -->
  <link rel="stylesheet" href="styles.css">

```

```

<!-- Internal CSS -->
<style>
  /* Internal CSS */
  header {
    background-color: #333;
    color: white;
    text-align: center;
    padding: 10px;
  }

  footer {
    background-color: #007BFF;
    color: white;
    text-align: center;
    padding: 10px;
    margin-top: 20px;
  }
  p.note {
    color: #008800;
  }
</style>
</head>
<body>
  <!-- Inline CSS -->
  <header>
    <h1 style="font-size: 2.5rem; font-weight: bold;">CSS Styling Demo</h1>
  </header>

  <div>
    <h2>Inline CSS Example</h2>
    <p style="color: red; font-weight: bold;">
      This paragraph is styled using inline CSS. Inline styles are defined directly in the HTML tag.
    </p>
  </div>

  <div>
    <h2>Internal CSS Example</h2>
    <p class="note">
      This paragraph is styled using internal CSS. Internal styles are defined in the <style> block
      within the <head> section.
    </p>
  </div>

  <div>
    <h2>External CSS Example</h2>
    <p class="abcd">
      This paragraph is styled using external CSS. External styles are defined in a separate CSS file
      and linked using the <link> tag.
    </p>
  </div>

  <footer>
    <p>&copy; 2024 CSS Styling Demo</p>
  </footer>

```

```
</body>
</html>
```

styles.css

```
/* External CSS */
```

```
body {
  font-family: Arial, sans-serif;
  margin: 20px;
  line-height: 1.6;
  background-color: #f9f9f9;
}
```

```
h1 {
  color: #007BFF;
  text-align: center;
}
```

```
p {
  color: #333;
  font-size: 1rem;
}
```

```
p.abcd {
  color: blue;
}
```

```
div {
  border: 1px solid #ddd;
  padding: 10px;
  background-color: #fff;
  margin-bottom: 10px;
}
```